HW1: Question Answering

[TA] Cheng-Tsung Lee (李承璁)

In this homework, we will explore the fascinating field of **Medical Question Answering** using LLMs. Specifically, we will focus on the domain of **Pathology**, where the goal is to predict the correct answer for multiple-choice questions (MCQs) by leveraging advanced prompting and finetuning techniques.

Dataset

The **PathoQA Dataset** consists of thousands of expert-annotated multiple-choice questions derived from pathology board exams and clinical case reports. We will work with this dataset to train and evaluate our generative models.

Tasks

1. **Data Preparation:**
   - ✧ Load and preprocess the PathoQA Dataset.
   - ✧ Split the data into training, validation, and testing sets.
   - ✧ Format the questions into structured prompts.
2. **Method Implementation:**
   - ✧ Design customized prompt template, the following strategies might help:
     1. Zero-Shot
     2. Few-Shot
     3. Chain-of-Thought (CoT)
   - ✧ Implement LoRA (Low-Rank Adaptation) finetuning on an open-source LLM.
3. **Training and Evaluation:**
   - ✧ Train the model using appropriate loss function.
   - ✧ Monitor training process and validate on the validation set.
   - ✧ Evaluate the model's performance using accuracy.
4. **Inference and Visualization:**
   - ✧ Apply the trained model to unseen sentences from the testing set.
   - ✧ Calculate the average accuracy throughout the testing set.
   - ✧ Visualize the outcome of both training and testing sets.

Learning Objectives

By completing this homework, we aim to:

- ✓ Gain hands-on experience with prompt engineering and efficient finetuning techniques.
- ✓ Learn to interpret question answering results and assess model performance.
- ✓ Understand the challenges of natural language processing.

Important Date:

1. Start Date: 2026/03/10 (Tue.) 16:00
2. Source code submission deadline: 2026/03/30 (Mon.) 23:59
   No late submission will be accepted afterwards.

Submission Form:

1. Please zip all files in one and named "HW1_{student_id}.zip" when uploading to E3 platform.

```
HW1_{student_id}/
├── saved_models/
├── main.py
├── README.md
└── requirements.txt
```

2. We will also have submission on "Kaggle". Make sure the team's name is your student ID, and the account is linked to the email provided on the class participant list. Each account can submit the answer 3 times every 24 hours.
3. Kaggle link: https://www.kaggle.com/t/68950a36b8cace8bed695c4679b1f8f7

Requirements:

1. Design your own prompting strategy.
2. Construct any training pipelines on your own.
3. You can ONLY use **`meta-llama/Llama-3.2-1B-Instruct`** as your model.
4. Plot each epoch's training phase and validating phase to observe how they change.

Implementation Details:

Dataset Files

The dataset includes two primary files:

`dataset.csv`: This file includes 9,001 csv rows, where there are a header and 9,000 rows of data represented as follows:

- question_id – the index of data
- question – sentence in string format
- opa, opb, opc, opd – potential options in string format
- ans – ground-truth correct answer in string format

`benchmark.csv`: This file includes 900 unseen questions without answer. This is used to measure the final performance of your models and training pipelines.

File Structure

Please organize your project files using the following directory structure that separates data, code, and other resources.

```
HW1_{student_id}/
├── dataset/
│   ├── benchmark.csv
│   └── dataset.csv
├── saved_models/
│   └── checkpoint
│       └── ...
├── main.py
├── README.md
└── requirements.txt
```

`dataset/`

This directory contains data related to the **PathoQA Dataset**. The dataset files (train, val, test.csv) are likely stored within this subdirectory. We access and manipulate the dataset from here.

`saved_models/`

This directory is where you would store trained model checkpoints. After training, the best-performing model are saved here for later use.

`main.py`

This file typically contains:

1. Utility functions used across the project. These functions might include visualization tools, metrics computations, or other common tasks.
2. Code for training the models. It includes functions related to model training and optimization.
3. Model evaluation. It includes functions to assess model performance on validation.

`README.md`

This file explains how to reproduce your final result. It probably includes the setting of virtual environment and training hyperparameters.

`requirements.txt`

This file lists all the packages (or libraries) used in the Python files.

Accuracy

The accuracy counts the number of correctly answered instances out of the total instances and is defined as the ratio of correct predictions to the total number of predictions.

$$Accuracy = \frac{\sum_{i=1}^{N} 1(\hat{y}_i = y_i)}{N}$$

Grading Policy:

Homework score = Experimental results (100%)

Experimental results (100%) (Note that result depends on benchmark set)

- [+15 pts]: accuracy @Top10%
- [+10 pts]: accuracy @Top10 ~ 25%
- [+5 pts]: accuracy @Top25 ~ 50%
- [85 pts]: accuracy ≥ 75%
- [70 pts]: accuracy ≥ 70%
- [0 pts]: accuracy < 70%

If the zip filename has a format error, it will be a penalty (-10 pts).

Please only train & validate your model on the training & validating dataset split by the provided dataset. If the TA found your model trained on the benchmark set or any other additional datasets through retraining your model, you'll get 0 pts in this homework.

Please ensure the TA can execute the code you uploaded. If the TA fails to do so, you'll get 0 pts in this homework.

If you're found faking your model predicting results, you'll get 0 pts in this homework. Also, we will take disciplinary action under school regulations.

Plagiarism is strictly prohibited. If you're found guilty of plagiarizing another's code or results, you and the person/people you plagiarized with will get 0 pts in this homework. Also, we will take disciplinary action under school regulations.

If any issues are identified in your submitted files for this lab, the TA will reach out to you via email. Please monitor your inbox closely. Suppose the TA has not received your response within three days of sending the email. In that case, it will be considered that you have forfeited the opportunity to provide further clarification and have accepted the corresponding penalty or consequences.

Contact & Information:

- Please post your question on the E3 forum.
- [TA] Cheng-Tsung Lee (李承璁): ctlee.ee14@nycu.edu.tw
- [TA hours] 11:00-13:00 Wed. ED-716 (Please make an appointment by email first)